Q1: Write MATLAB code to do the following:

1. **Read the color image**
2. **Display the color image**
3. **Draw the histogram of the color image**
4. **Calculate Mean Absolute Error(MAE)**
5. **Calculate Mean Squared Error(MSE)**
6. **Calculate Peak Signal To Noise Ratio (PSNR)**
7. **Get the negative of the color image and histogram**
8. **Get the Log of the color image and histogram**
9. **Get the Power Law (Gamma) of the color image and histogram**
10. **Get the Thresholding of the color image and histogram**
11. **Get the Fixed Intensity transformation of the color image and histogram**
12. **Get the Full Scale Contrast Stretch of the color image and histogram**
13. **Get the grey image for the color image**

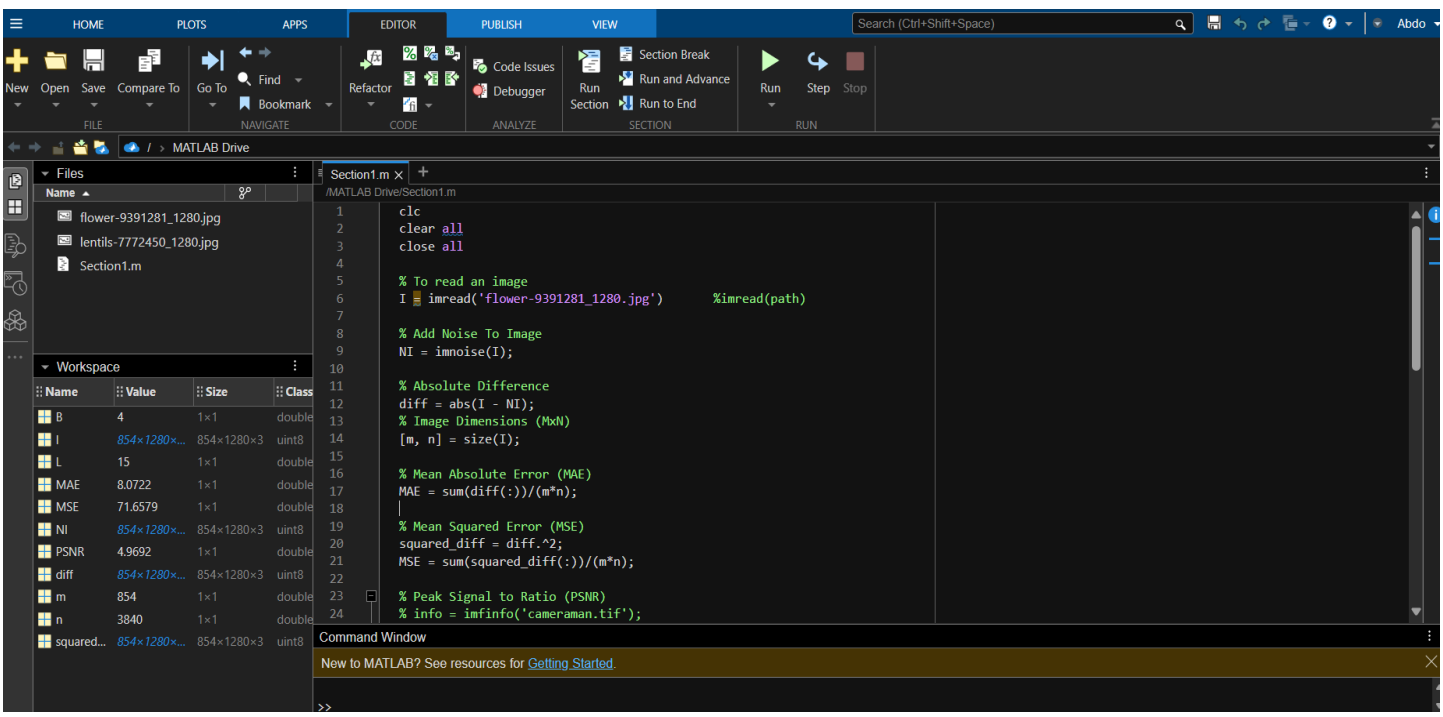
Solution :

MATLAB Code :

Image Quality Measures

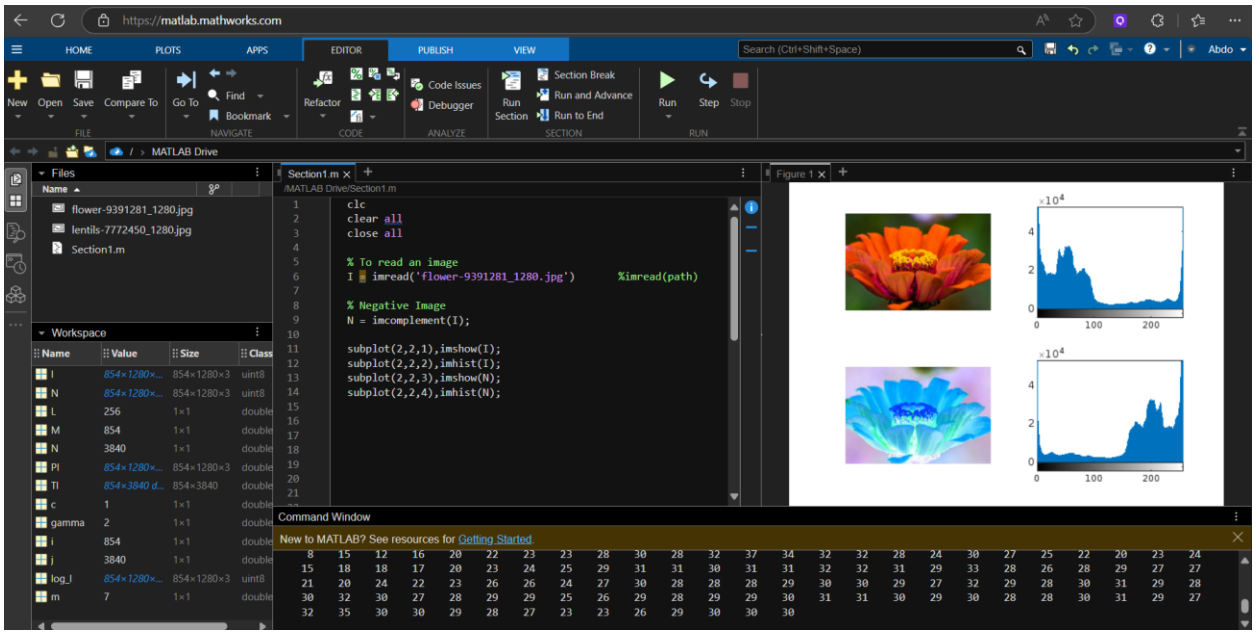
```
clc
clear all
close all

% To read an image
I = imread('flower-9391281_1280.jpg')                    %imread(path)

% Add Noise To Image
NI = imnoise(I);

% Absolute Difference
diff = abs(I - NI);
% Image Dimensions (MxN)
[m, n] = size(I);

% Mean Absolute Error (MAE)
MAE = sum(diff(:))/(m*n);

% Mean Squared Error (MSE)
squared_diff = diff.^2;
```

```
MSE = sum(squared_diff(:))/(m*n);


% Peak Signal to Ratio (PSNR)
% info = imfinfo('cameraman.tif');
% B = info.BitDepth;
B = 4;
L = 2.^B - 1;
PSNR = 10*log10((L.^2)/MSE);
```

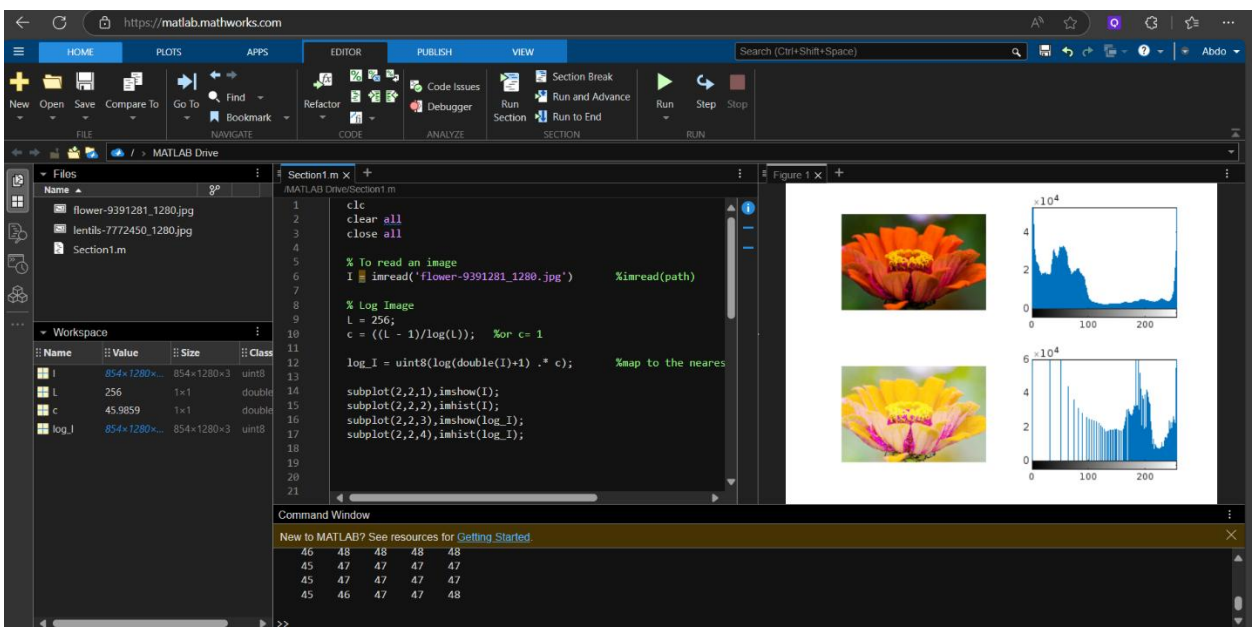


Intensity Transformations

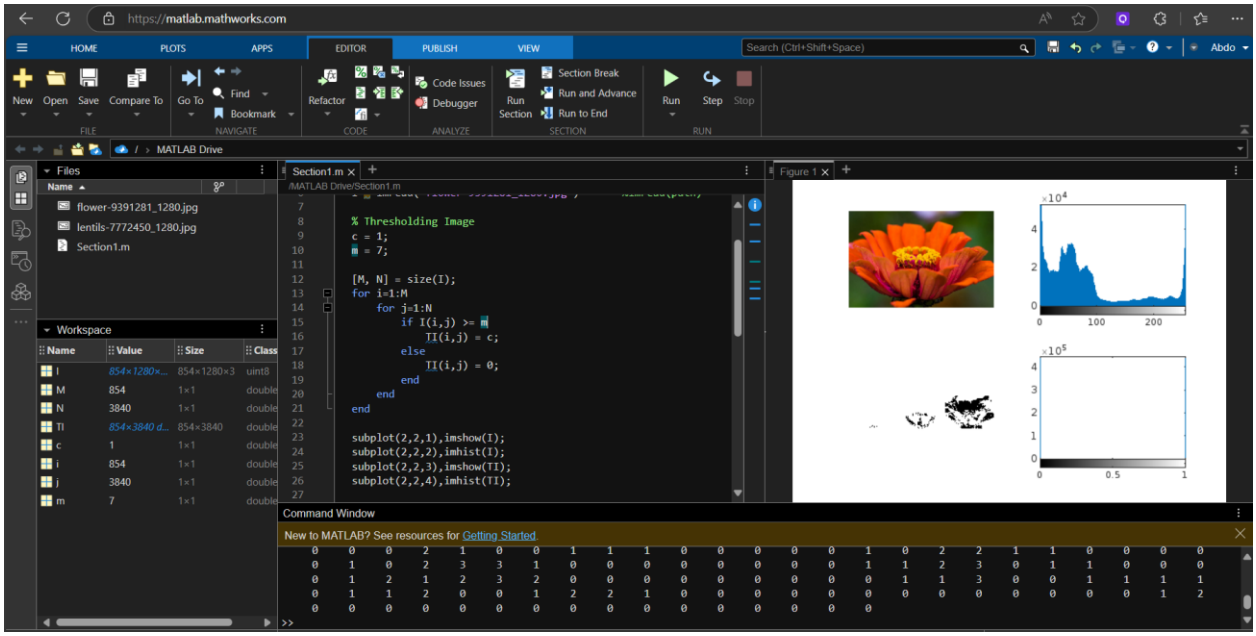
1. Negative



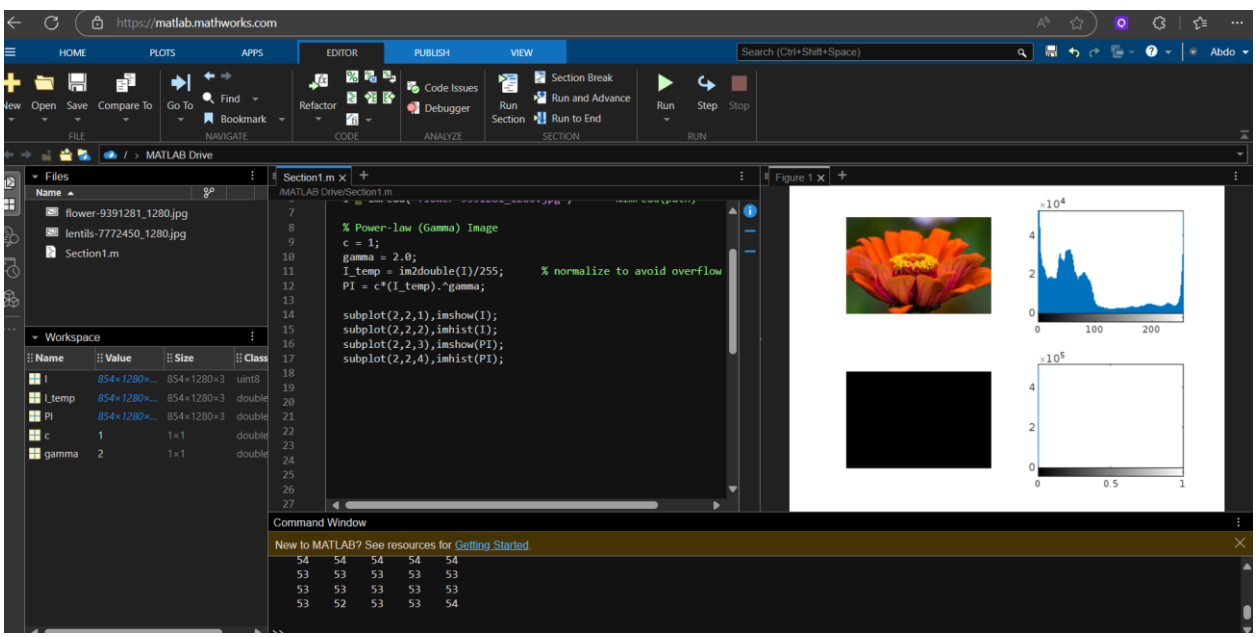
2. Log



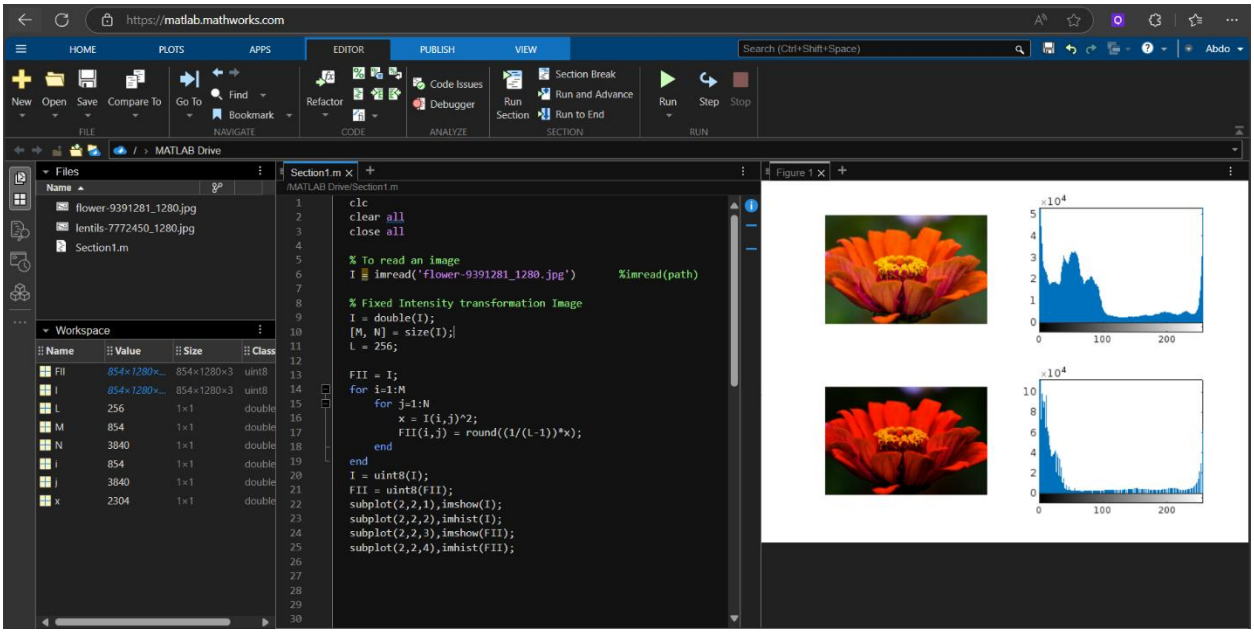
3. Thresholding



4. Power-law (Gamma)



5. Fixed Intensity Transformation



6. Full Scale Contrast Stretch

```
clc
clear all
close all

% To read an image
I = imread('flower-9391281_1280.jpg')          %imread(path)

% Full Scale Contrast Stretch  Image
FSI = imadjust(I);
```

```
subplot(2,2,1),imshow(I);
subplot(2,2,2),imhist(I);
subplot(2,2,3),imshow(FSI);
subplot(2,2,4),imhist(FSI);
```